

BCA-5003: COMPUTER NETWORK

UNIT III — COMPLETE DETAILED NOTES + TOPIC-WISE PYQ ANSWER BOOK

Telephony • Point-to-Point Controls • ISDN • Switching • Data-Link Control • Multiplexing

Unit-I format: Index → detailed notes → diagrams/graphs → PYQs with years → long model answers → revision plan

Prepared from the supplied Unit-III content. Technical inaccuracies in raw material have been corrected for safe exam writing.

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4	Sliding-window protocol, Go-Back-N and Selective Repeat ARQ	2021, 2023–24, 2025–26
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How to score well in Unit III

1. Memorize all three switching techniques with diagrams and comparison table.
2. Practise Stop-and-Wait efficiency and learn what propagation delay and transmission time mean.
3. Draw Go-Back-N and Selective Repeat sequence diagrams until you can reproduce them without looking.
4. For multiplexing, draw MUX → shared link → DEMUX and explain FDM, TDM and WDM in separate paragraphs.
5. For ISDN, always include B, D and H channels plus bearer, teleservices and supplementary services.

1. Unit-III Roadmap

Unit III focuses on reliable data transfer, switching, multiplexing, telephony/ISDN, and congestion control. It connects the Data-Link Layer with wide-area networking concepts. The most scoring parts are diagrams, comparison tables, flow-control protocols, switching techniques, multiplexing, and ISDN.

Recommended order

1. Data-link layer functions and framing.
2. Stop-and-Wait, piggybacking, sliding window, Go-Back-N and Selective Repeat.
3. Circuit, message and packet switching.
4. Multiplexing: FDM, TDM, WDM.
5. ISDN architecture, channels and services.
6. Congestion and traffic shaping: leaky bucket and token bucket.

Exam writing rule

Whenever a question says “explain”, write an introduction, a labelled diagram, working steps, advantages/limitations, and a conclusion. For comparison questions, prefer a table. For protocol questions, show sender and receiver interaction using arrows and time direction.

2. Data-Link Layer: Services and Core Functions

The Data-Link Layer is Layer 2 of the OSI model. It provides node-to-node delivery over a single communication link. While the Physical Layer sends raw bits, the Data-Link Layer groups those bits into frames and manages local delivery, errors, flow, and access to a shared medium.

Core functions

Function	Detailed meaning
Framing	Divides continuous bit stream into identifiable blocks called frames. Headers/trailers carry addressing, control, and error-detection information.
Physical/MAC addressing	Adds local source and destination hardware addresses for link-level delivery.
Error control	Detects damaged/lost/duplicate frames using CRC/checksums, ACK/NAK, sequence numbers, timers, and retransmission.
Flow control	Prevents a fast sender from overflowing a slow receiver's buffer or processing capacity.
Access control	Coordinates which device may transmit when several devices share a common medium.
Link management	Establishes, maintains, and releases a link when connection-oriented operation is used.

Frame concept diagram

Raw bit stream from Physical Layer:
101011001001101010010101001010...

Data-Link Layer groups it as:
[Header | Payload/Data | Trailer/FCS] [Header | Payload/Data | Trailer/FCS]

A frame header may contain MAC addresses, sequence/control information, and a type/length field. A trailer commonly carries error-detection bits such as Frame Check Sequence (FCS).

3. Framing

Framing is the Data-Link Layer process of dividing a continuous stream of bits into meaningful, manageable units called frames. The receiver must be able to identify where each frame begins and ends. Frames make addressing, error checking, flow control, and retransmission possible.

Why framing is necessary

- The Physical Layer provides an unstructured bit stream; the Data-Link Layer needs clear boundaries.

- A receiver must know which bits belong together for CRC/error detection.
- Lost or damaged units can be retransmitted as individual frames instead of retransmitting a complete file.
- Sequence numbers and acknowledgements operate on frames.

Common framing approaches

Method	Idea	Example
Character/byte count	Header carries length of the frame.	Some legacy protocols.
Byte stuffing	Special flag bytes mark boundaries; escape bytes handle flag-like data.	PPP-style byte-oriented framing.
Bit stuffing	Special flag bit pattern marks frames; sender inserts a 0 after five consecutive 1s in data.	HDLC.
Physical-layer coding violation	Uses an unused physical signal pattern to mark boundary.	Certain LAN designs.

Typical frame formats

Ethernet (conceptual):

[Preamble | SFD | Destination MAC | Source MAC | Type/Length | Data + Pad | FCS]

PPP:

[Flag | Address | Control | Protocol | Payload | FCS | Flag]

HDLC:

[Flag | Address | Control | Information | FCS | Flag]

ATM cell:

[5-byte Header | 48-byte Payload] – fixed 53-byte cell.

4. Error Control, Stop-and-Wait and Piggybacking

Error control is the set of procedures that ensures frames are delivered correctly despite loss, corruption, duplication, or acknowledgement loss. It uses error detection, ACK/NAK messages, sequence numbers, timers, and retransmission.

Error-control building blocks

- Error-detection code: CRC/checksum/parity detects corruption.
- Sequence numbers: distinguish new frames from duplicate retransmissions.
- Acknowledgement (ACK): confirms a correct frame.
- Negative acknowledgement (NAK): requests a specific retransmission where used.
- Timer: prevents sender from waiting forever if a frame/ACK is lost.

Stop-and-Wait ARQ

In Stop-and-Wait ARQ, the sender sends one frame and waits for its acknowledgement before sending the next frame. Sequence numbers (usually 0 and 1) prevent duplicate-delivery errors after retransmission. The method is simple but has poor utilization when propagation delay is high compared with frame transmission time.

Stop-and-Wait time diagram



Efficiency of Stop-and-Wait (ideal simplified case)

Let T_t = frame transmission time and T_p = one-way propagation delay. Ignoring ACK transmission/processing delay:

Efficiency $\eta = T_t / (T_t + 2T_p) = 1 / (1 + 2a)$, where $a = T_p / T_t$.

Piggybacking

In full-duplex communication, both sides may send data. Instead of sending a separate ACK frame immediately, a receiver can delay the ACK briefly and place it inside the header of its next outgoing data frame. This is called piggybacking: the acknowledgement gets a “free ride” with reverse-direction data.

Piggybacking diagram

```
A → B: [Data frame 0]
B → A: [Data frame 0 + ACK for A's frame] ← piggybacked ACK
A → B: [Data frame 1 + ACK for B's frame]
```

Benefit: fewer separate control frames, lower overhead, better bandwidth use. Limitation: ACK cannot wait indefinitely; a delayed-ACK timer must send a separate ACK if reverse-direction data does not arrive soon.

5. Sliding Window, Go-Back-N and Selective Repeat

Sliding-window protocols improve link utilization by allowing multiple frames to be sent before acknowledgements are received. The sender maintains a window of sequence numbers it is permitted to transmit. As ACKs arrive, the window advances—or “slides”—and additional frames may be sent.

Sliding-window diagram

```
Sequence numbers: 0 1 2 3 4 5 6 7
Sender window: [0] [1] [2] [3] 4 5 6 7
After ACK for 0: 0 [1] [2] [3] [4] 5 6 7
```

Go-Back-N ARQ

Go-Back-N allows the sender to transmit up to N unacknowledged frames. The receiver normally accepts only the next expected frame; its receive window is effectively one. If frame k is lost or damaged, frames after k are discarded even if they arrive correctly. The sender retransmits frame k and all later outstanding frames after a NAK or timeout.

Go-Back-N loss diagram

Sender	Receiver
Frame 0	accepted; ACK 1
Frame 1	accepted; ACK 2
Frame 2	X lost/damaged
Frame 3	discarded (expects 2)
Frame 4	discarded (expects 2)
	NAK 2 / repeated ACK 2
Frame 2	retransmitted
Frame 3	retransmitted
Frame 4	retransmitted

Selective Repeat ARQ

Selective Repeat allows the receiver to accept and buffer correctly received frames that arrive out of order. If frame k is lost/damaged, the sender retransmits only frame k. The receiver then places buffered frames in sequence once the missing frame is obtained. This is more bandwidth-efficient than Go-Back-N but needs more complex sender/receiver logic and buffer space.

Selective Repeat loss diagram

Sender	Receiver
Frame 0	accepted
Frame 1	accepted
Frame 2	X lost/damaged
Frame 3	buffered (out of order)
Frame 4	buffered (out of order)
	NAK 2
Frame 2	retransmitted; then 2,3,4 delivered in order

Go-Back-N versus Selective Repeat

Basis	Go-Back-N	Selective Repeat
Retransmission	Resends lost/damaged frame and all subsequent outstanding frames.	Resends only lost/damaged frame(s).
Receiver behaviour	Discards out-of-order frames.	Buffers valid out-of-order frames.
Complexity	Lower.	Higher due to buffers, individual ACKs/timers, sorting.

Basis	Go-Back-N	Selective Repeat
Bandwidth when errors occur	May waste bandwidth by retransmitting good frames.	More efficient; avoids unnecessary retransmission.
Receiver window	Usually 1.	More than 1; must be limited to avoid sequence ambiguity.

6. Switching Techniques

Switching is the process of transferring data from a source to a destination through intermediate network nodes. Switching techniques decide whether a route is reserved, whether an entire message is stored, or whether data is divided into packets. The three classical techniques are circuit switching, message switching, and packet switching.

Circuit switching

Circuit switching establishes a dedicated end-to-end path before data transfer. Resources along the path are reserved for the complete session. It has three phases: setup, data transfer, and teardown. Traditional telephone networks are the standard example.

Circuit-switching phases

```
A ■■ setup request ■■■ [Switch 1] ■■■ [Switch 2] ■■■ B
A ■■■■■■■■■■ dedicated path established ■■■■■■■■■■ B
A ■■■■■■■■■■ reserved circuit/data flow ■■■■■■■■■■ B
A ■■■■■■■■■■ teardown / release ■■■■■■■■■■ B
```

Advantages: predictable bandwidth/delay after setup, in-order delivery, suitable for continuous real-time streams.

Limitations: idle reserved bandwidth is wasted; setup required; a path failure interrupts the circuit.

Message switching

Message switching transfers the entire message from one node to the next using store-and-forward operation. No dedicated path is required. Every intermediate node must receive and store the complete message before forwarding it. This can produce high delay and requires large buffers. Telegraph systems historically used this approach.

Message-switching diagram

```
Source ■■ [store entire M] ■■ Switch 1 ■■ [store entire M] ■■ Switch 2 ■■ Destination
```

Packet switching

Packet switching divides a message into smaller packets. Each packet carries control/addressing information and is forwarded through the network. At the destination, packets are reassembled. Packet switching uses links efficiently because multiple users can share capacity statistically. The Internet is based on packet switching.

Packet-switching diagram

```
Message → [P1][P2][P3][P4]
P1 → Route A → Destination
P2 → Route B → Destination
P3 → Route A → Destination
P4 → Route C → Destination
Destination reassembles P1,P2,P3,P4 in correct order.
```

Packet switching types

Type	Working	Example
Virtual circuit / connection-oriented	A logical path is set up first; packets follow that path and use a virtual-circuit identifier.	X.25, Frame Relay, ATM, MPLS concepts.
Datagram / connectionless	No setup; every packet is independently routed and carries destination address. Packets may follow different paths/arrive out of order.	Internet Protocol (IP).

Comparison of switching techniques

Basis	Circuit switching	Message switching	Packet switching
Unit transferred	Continuous stream over reserved circuit.	Complete message.	Small packets.
Setup	Required.	Not required.	Datagram: no; virtual circuit: yes.
Resource reservation	Yes.	No.	Usually no dedicated physical reservation.
Store-and-forward	No for each message; path pre-established.	Entire message stored at each node.	Usually packet stored/forwarded at each node.

Basis	FDM	TDM	WDM
Transmission	All channels simultaneously at different frequencies.	Channels take turns in time.	All optical channels simultaneously at different wavelengths.
Need	Guard bands/filtering.	Synchronization and framing.	Optical MUX/DEMUX, wavelength control.
Example	Radio broadcasting.	Digital telephony/links.	Fibre-optic backbone.

8. ISDN: Integrated Services Digital Network

ISDN is a digital telecommunications technology designed to carry voice, data, text, images, and signalling over a common digital network interface. It was historically important because it integrated multiple services on telephone infrastructure and allowed simultaneous voice/data communication.

ISDN channels

Channel	Typical rate	Purpose
B channel (Bearer)	64 kbps	Carries user information such as voice, data, fax, or other bearer traffic. Multiple B channels may be bonded to increase the rate.
D channel (Delta/Data signalling)	16 kbps in BRI; 64 kbps in PRI	Primarily carries signalling/control information: call setup, call clearing, ringing, and network control. It may also carry low-speed data.
H channel (High-rate)	H0 = 384 kbps; H11 = 1.536 Mbps; H12 = 1.920 Mbps	Carries high-speed information such as video, high-speed data, or multiple combined channels.

ISDN access types

Interface	Composition	Typical context
BRI — Basic Rate Interface	2B + D = two 64-kbps B channels plus 16-kbps D channel.	Homes/small offices; total user channels 144 kbps excluding D channel.
PRI — Primary Rate Interface	T1 regions: 23B + D; E1 regions: 30B + D.	Larger organisations and PBX connections.

Conceptual ISDN protocol architecture

OSI / ISDN concept:

Network layer: Call control (Q.931) / packet services as applicable

Data-link layer: LAPD (D channel signalling); user data protocols on B channels

Physical layer: BRI / PRI interface, line coding, electrical/mechanical specifications

B channels: primarily user bearer information

D channel: signalling and control

ISDN services

Service category	Meaning	Examples
Bearer services	Network transports information without changing/manipulating its content; broadcast and interactive, packet or circuit-switched.	Any real-time data, QoS key data transport.
Teleservices	Complete end-user communication services; network/application functions may process context.	Telephone, confer, teletex, videotex, teleconferencing.
Supplementary services	Additional capabilities added to bearer/teleservices.	Call waiting, caller ID, call forwarding, conference calling, reverse charges.

Narrowband ISDN versus Broadband ISDN

Basis	N-ISDN	B-ISDN
Basic unit / rate	Built around 64-kbps channels.	Designed for much higher rates, typically broadband services.
Transport orientation	Historically circuit switching and packet/frame services.	Broadband packet/cell-based approaches; ATM historically central.
Medium	Telephone copper and digital telephony infrastructure.	Fibre-optic and broadband-capable media.
Typical services	Voice, low-rate data, fax.	High-speed data, video, multimedia.

9. Congestion Control and Traffic Shaping

Congestion occurs when packets arrive at routers or network links faster than they can be processed, queued, and forwarded. Queues grow, delay rises, buffers overflow, packets are dropped, and retransmissions may create even more traffic. In severe cases, the network can experience congestion collapse.

Congestion feedback cycle

More offered traffic → queues grow → delay/loss increases → senders retransmit → even more traffic → congestion worsens

Congestion control aims to regulate the entry and rate of traffic so that the network can operate efficiently. Traffic shaping is one approach: it controls the rate and burst pattern before packets enter the network.

Leaky bucket algorithm

Leaky bucket shapes bursty traffic into a steady output stream. Incoming packets enter a finite queue/bucket. Packets leave at a fixed rate. If the bucket/queue becomes full, newly arriving packets are discarded or marked depending on implementation.

Leaky-bucket diagram

Bursty packets → [Bucket / finite queue] → fixed-rate leak → Network
overflow when full → packet drop

Advantages: smooth, predictable output; protects network from sudden bursts.

Limitation: it does not allow a host to use previously unused capacity to transmit a later burst; excess packets can be dropped when queue is full.

Token bucket algorithm

Token bucket controls average traffic rate while allowing controlled bursts. Tokens are added to a bucket at a fixed rate up to a maximum capacity. A packet may be sent only when sufficient tokens are available; sending consumes tokens. If the host has been idle, tokens accumulate and it can send a burst up to the stored-token limit.

Token-bucket diagram

Clock → token generator → [Token bucket (max capacity)]
Packets [Need token(s)?] ■ yes → transmit to network
■■ no → wait or queue/drop (policy dependent)

Leaky bucket versus token bucket

Basis	Leaky bucket	Token bucket
Output rate	Usually fixed/constant.	Average rate controlled; bursts allowed if tokens accumulated.
Idle-period credit	No useful credit accumulates for future bursts.	Tokens accumulate up to bucket capacity.
Burst handling	Smooths/removes bursts; overflow may be dropped.	Allows bursts within stored token limit.
Best use	Strict rate shaping.	Rate limiting with controlled burst tolerance.

10. Applications of Computer Networks

Computer networks support communication, resource sharing, remote access, distributed applications, Internet services, and collaboration. They are fundamental to education, banking, health care, e-commerce, government services, cloud computing, and software development.

- **Data sharing:** multiple authorised users can access databases, documents, and shared information.
- **Remote data access:** users can securely access applications/files from another location through cloud services, VPNs, or remote desktop.
- **Resource sharing:** printers, storage, servers, software licences, and Internet links can be shared.
- **Communication:** email, chat, VoIP, video meetings, social platforms, and collaboration tools.
- **Internet:** access to web services, online learning, streaming, search, e-commerce, and digital government.
- **Intranet:** private internal organisational network for employees and internal applications.
- **Extranet:** secure extension of an intranet for authorised external partners, suppliers, or customers.
- **File sharing and teamwork:** cloud storage and collaborative editing support team projects in real time.

11. Topic-Wise PYQ Answer Bank

The questions below are mapped to the Unit-III concepts above. Each model answer is written for detailed university examination preparation.

Q1. What is piggybacking?

Asked: 2023–24, 2025–26

Answer

Piggybacking is an acknowledgement technique used in full-duplex data communication. When both stations have data to send, a receiver does not immediately transmit a separate ACK frame for every incoming data frame. Instead, it waits briefly and attaches the acknowledgement number to the header of its next outgoing data frame.

In this way, the ACK gets a “free ride” with the reverse-direction data frame. A separate ACK would otherwise need its own header, control information, and error-checking field, creating unnecessary overhead.

Working: Station A sends a data frame to station B. B receives it correctly and must acknowledge it. If B has its own data ready to send to A, B includes the ACK field in that outgoing data frame. If B has no outgoing data within a short delay interval, B sends a separate acknowledgement so that A is not forced to wait unnecessarily.

Advantages: reduces control-frame overhead, uses bandwidth more efficiently, and is especially useful in bidirectional full-duplex data transfer.

Limitation: the receiver must not delay an ACK indefinitely; a delayed-ACK timer is necessary.

Conclusion: Piggybacking improves efficiency by combining acknowledgement and reverse-direction data in one frame.

Q2. A channel has bit rate 4 kbps and one-way propagation delay 20 ms. For what minimum frame size does Stop-and-Wait give efficiency at least 50%?

Asked: 2017

Answer

For Stop-and-Wait ARQ, ignoring ACK transmission and processing delays:

$$\eta = T_t / (T_t + 2T_p)$$

Given: $\eta = 0.5$, bit rate $R = 4 \text{ kbps} = 4000 \text{ bps}$, one-way propagation delay $T_p = 20 \text{ ms} = 0.020 \text{ s}$.

$$0.5 = T_t / (T_t + 2 \times 0.020)$$

$$0.5(T_t + 0.040) = T_t$$

$$0.5T_t + 0.020 = T_t$$

$$0.5T_t = 0.020$$

$$T_t = 0.040 \text{ second} = 40 \text{ ms}$$

Frame size = bit rate $\times$ transmission time

$$= 4000 \text{ bits/s} \times 0.040 \text{ s}$$

= **160 bits.**

Answer: The frame size must be **at least 160 bits** for efficiency of at least 50%. Any frame size greater than or equal to 160 bits satisfies the condition under the stated assumptions.

Q3. Describe the sliding-window protocol. / Write short note on Go-Back-N.

Asked: 2021, 2023–24, 2025–26

Answer

The sliding-window protocol is a flow-control and error-control technique that permits the sender to transmit multiple frames before waiting for acknowledgements. The number of unacknowledged frames that may be sent is the sender window size. When ACKs arrive, the sending window slides forward and new frames become eligible for transmission.

It is more efficient than Stop-and-Wait because the link can remain busy while earlier frames are propagating and acknowledgements are returning. Sliding-window protocols use sequence numbers to identify frames.

Go-Back-N ARQ: The sender may have up to N outstanding frames. The receiver accepts only the next expected frame and discards later out-of-order frames. If frame 2 is lost, frames 3 and 4 are discarded even if they arrive correctly. The sender retransmits frame 2 and every later unacknowledged frame after a timeout or NAK.

Advantages: higher throughput than Stop-and-Wait; sender implementation is simpler than Selective Repeat.

Disadvantage: when errors occur, correctly transmitted later frames may be unnecessarily retransmitted, wasting bandwidth.

Conclusion: Sliding-window protocols improve performance by pipelining frames; Go-Back-N provides a practical trade-off between simplicity and efficiency.

Q4. Compare Go-Back-N and Selective Repeat.

Asked: 2021

Answer

Go-Back-N: retransmits the damaged/lost frame and all subsequent outstanding frames. The receiver discards out-of-order frames and normally has a receive window of one. It is simpler but wastes bandwidth when an error occurs.

Selective Repeat: retransmits only the specific damaged/lost frame. The receiver accepts/buffers valid out-of-order frames, requiring larger buffers and more complex control. It uses bandwidth more efficiently, especially on links with non-trivial error rate.

Key comparison: Go-Back-N has low complexity and potentially high retransmission waste; Selective Repeat has higher complexity but avoids retransmitting frames already received correctly.

Q5. Explain different types of switching techniques.

Asked: 2024–25; related: 2019, 2022–23, 2025–26

Answer

Switching is the process of moving information through intermediate network nodes from source to destination. The principal switching techniques are circuit switching, message switching, and packet switching.

1. Circuit switching: a dedicated end-to-end path is established before transmission. Resources are reserved for the complete session. It has setup, data-transfer, and teardown phases. It offers predictable performance after setup but wastes reserved capacity during silence. Traditional telephone calls are an example.

2. Message switching: the complete message is stored at every intermediate node and then forwarded when the next link is available. There is no dedicated path. It needs large buffers and can create high delay, especially for long messages. Telegraph networks are a historical example.

3. Packet switching: the message is divided into small packets. Packets are stored and forwarded individually, allowing better sharing of network links. In datagram packet switching, every packet is routed independently; in virtual-circuit packet switching, a logical route is established first. The Internet uses packet switching.

Conclusion: Circuit switching is best for a dedicated continuous path; message switching is mainly historical due to high delay; packet switching is the dominant technique in modern data networks because of efficient resource sharing.

Q6. What is framing? Explain different framing protocols with formats.

Asked: 2023–24, 2024–25, 2025–26

Answer

Framing is a Data-Link Layer function that divides a continuous bit stream into identifiable blocks called frames. A frame contains a header, payload, and often a trailer. It permits error detection, addressing, sequencing, flow control, and reliable retransmission.

Ethernet frame: [Preamble | Start Frame Delimiter | Destination MAC | Source MAC | Type/Length | Data/Pad | FCS]. It is used in LAN Ethernet networks. The FCS commonly uses CRC for error detection.

PPP frame: [Flag | Address | Control | Protocol | Payload | FCS | Flag]. PPP is used on point-to-point links and supports link setup/control functions.

HDLC frame: [Flag | Address | Control | Information | FCS | Flag]. HDLC is a bit-oriented synchronous Data-Link protocol for point-to-point/multipoint links. It uses flag sequences and bit stuffing.

Frame Relay: uses variable-length frames and virtual circuits in WANs; its header identifies the virtual connection, commonly through DLCI-related information.

ATM: uses fixed 53-byte cells: [5-byte header | 48-byte payload]. It was designed for efficient integrated voice, video, and data transport using virtual circuits.

SONET/SDH: optical telecom transport standards that use synchronised frame structures with overhead and payload sections for high-speed network transport.

Conclusion: Framing protocols define how devices identify frame boundaries and interpret addressing, control, payload, and error-checking fields.

Q7. Explain multiplexing. Explain FDM, TDM and WDM.

Asked: 2017, 2021, 2022–23, 2023–24

Answer

Multiplexing is a method of combining several independent signals/data streams so that they can share one high-capacity communication link. A multiplexer combines the streams at sender side, and a demultiplexer separates them at receiver side.

FDM: Frequency-Division Multiplexing divides link bandwidth into separate frequency bands. Each signal modulates a different carrier and all channels travel simultaneously. Guard bands reduce overlap. FDM is mainly associated with analog/RF communication, such as radio broadcasting.

TDM: Time-Division Multiplexing divides access by time. Each source gets a time slot and uses the full channel bandwidth during that slot. Synchronous TDM reserves slots even for idle sources; statistical TDM assigns slots dynamically to active sources. TDM is widely used for digital data.

WDM: Wavelength-Division Multiplexing combines light signals of different wavelengths on one optical fibre. It is the fibre-optic counterpart of FDM. DWDM allows many closely spaced optical channels on the same fibre.

Conclusion: FDM shares frequencies, TDM shares time, and WDM shares optical wavelengths. All improve utilisation of a high-capacity link.

Q8. Explain ISDN architecture and its services.

Asked: 2023–24, 2025–26; related: 2015, 2018, 2021, 2022–23

Answer

ISDN stands for Integrated Services Digital Network. It is a digital telecommunication network designed to carry voice, data, text, images, and signalling services through standard digital interfaces. It can support simultaneous voice and data communication.

Channels: B channel is a 64-kbps bearer channel that carries user voice/data. D channel is mainly for signalling and control; it is 16 kbps in BRI and 64 kbps in PRI. H channels provide higher rates: H0 = 384 kbps, H11 = 1.536 Mbps, and H12 = 1.920 Mbps.

Interfaces: BRI = 2B + D, mainly for smaller users. PRI provides many B channels plus one D channel—23B+D in T1 systems and 30B+D in E1 systems.

Architecture: Physical layer defines BRI/PRI interfaces and line/electrical characteristics. D-channel data-link signalling uses LAPD. Call control is handled by protocols such as Q.931. B channels carry user bearer traffic and may connect to circuit-switched, packet-switched, or other networks.

Services: Bearer services transfer information without altering content. Teleservices provide end-user services such as telephony, fax, teletex, videotex, and teleconferencing. Supplementary services add functions such as call waiting, caller identification, call forwarding, and conference calling.

Conclusion: ISDN integrated several digital telecommunication services over common interfaces and was important in the transition from analog telephony to modern digital networks.

Q9. Describe congestion control algorithms. Explain leaky bucket and token bucket.

Asked: 2023–24, 2024–25, 2025–26

Answer

Congestion occurs when traffic offered to a network exceeds its ability to process and forward packets. Queues become long, delay increases, buffers overflow, packets are dropped, and retransmissions can make the situation worse. Congestion control prevents or reduces this overload.

Leaky bucket algorithm: Incoming packets are placed in a finite queue/bucket. Packets are released at a constant rate, like water leaking through a small hole. Bursty input is converted into smooth output. If the bucket becomes full, excess packets are discarded or marked. It provides strict rate shaping but does not allow a later burst to use capacity saved during an idle period.

Token bucket algorithm: Tokens are generated at a fixed rate and stored in a bucket up to a maximum. Sending a packet requires enough tokens; sending consumes tokens. During idle time, tokens accumulate, so a source may later send a burst up to the available token amount. It controls average rate while allowing controlled bursts.

Comparison: Leaky bucket gives a fixed smooth output; token bucket permits bursts after idle periods. Leaky bucket is strict shaping; token bucket is flexible rate limiting.

Conclusion: Both mechanisms regulate traffic before it overloads network resources, but token bucket is usually preferred where burst tolerance is required.

Q10. Explain applications of computer networks.

Asked: 2023–24, 2025–26

Answer

Computer networks connect devices so that users and organisations can share information, communicate, access remote resources, and run distributed services.

- 1. Data sharing:** users share files, databases, documents, and information through servers/cloud storage.
- 2. Remote data access:** authorised users access data and applications from remote locations using Internet, VPN, and remote desktop services.
- 3. Resource sharing:** shared printers, storage, Internet access, servers, and software reduce cost.
- 4. Communication:** email, messaging, VoIP, video conferencing, online classes, and collaborative platforms enable rapid interaction.
- 5. Internet services:** web browsing, e-commerce, digital payments, social media, streaming, search, and cloud applications.
- 6. Intranet and extranet:** intranets support internal organisational work; extranets provide controlled access to suppliers, partners, or customers.
- 7. Collaboration:** teams can co-edit documents, manage projects, exchange code, and work across locations.

Conclusion: Networks are the infrastructure behind modern communication, business operations, education, cloud computing, and digital services.

12. Final Revision Sheet

Must-draw diagrams

- Stop-and-Wait sender/receiver timing diagram.
- Piggybacked acknowledgement diagram.
- Sliding-window sequence-number diagram.
- Go-Back-N and Selective Repeat loss/retransmission diagrams.
- Circuit switching: setup → data transfer → teardown.
- Message switching: store-and-forward path.
- Packet switching: packets using different routes.
- MUX → shared link → DEMUX; FDM frequency bands; TDM time slots; WDM wavelengths.
- ISDN B/D/H channel architecture and satellite-like channel mapping is not required; draw correct ISDN interface diagram.
- Leaky bucket and token bucket diagrams.

Formula box

Stop-and-Wait efficiency: $\eta = T_t / (T_t + 2T_p) = 1/(1 + 2a)$

Where: T_t = frame transmission time, T_p = one-way propagation delay, $a = T_p/T_t$.

Frame size: Frame size in bits = link bit rate $\times T_t$.

Fast comparisons to memorise

Topic	One-line difference
Stop-and-Wait vs Sliding Window	Stop-and-Wait allows one outstanding frame; sliding window allows multiple outstanding frames.
Go-Back-N vs Selective Repeat	GBN resends from error onward; SR resends only erroneous/lost frame and buffers others.
Circuit vs Packet switching	Circuit reserves an end-to-end path; packet switching shares links using packets.
FDM vs TDM	FDM divides frequency; TDM divides time.
TDM vs WDM	TDM uses time slots for data streams; WDM uses different light wavelengths in fibre.
Leaky vs Token bucket	Leaky bucket makes fixed-rate output; token bucket allows controlled bursts.
BRI vs PRI	BRI is 2B+D; PRI has many B channels plus one D channel.

7-Day Unit-III Study Plan

Day	Focus	Output
1	Data-link functions and framing	Write frame definition and formats; practise Q6.
2	Stop-and-Wait, piggybacking, efficiency	Solve the 2017 numerical repeatedly; write Q1/Q2.
3	Sliding window, GBN, SR	Draw time diagrams; learn Q3/Q4.
4	Switching	Draw all three switching techniques; write Q5.
5	Multiplexing	Draw FDM/TDM/WDM and learn comparison; write Q7.
6	ISDN and congestion control	Prepare Q8/Q9 including diagrams and tables.
7	Full PYQ mock	Write Q1–Q10; revise diagrams/formula sheet.

Final accuracy notes

- A D channel is primarily a signalling/control channel; do not state that it “does not carry data” in every possible context.
- Go-Back-N receiver normally discards out-of-order frames; Selective Repeat buffers them.
- Message switching stores the entire message at each node; packet switching normally stores/forwards packets.
- FDM is frequency sharing, TDM is time sharing, WDM is optical wavelength sharing.

- Token bucket permits a burst only if enough tokens have accumulated; it does not allow unlimited bursting.
- ISDN is historically important; use its classical BRI/PRI terminology accurately in exams.

For a 10-CGPA attempt: do active recall. Close the notes, draw the diagrams and write each PYQ in handwriting; then check only what you missed. Repeat at least twice before the exam.